

STAT 312: Midterm Exam 01 Review

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Question 1 - Stem-and-Leaf Plots

Biologists are studying a woman who has given birth to twenty children (ten boys and ten girls). When each child reached 10 years of age, their height (in inches) was recorded:

Boys:

51.3, 52.8, 53.4, 53.5, 54.1, 54.6, 54.6, 55.2, 55.7, 55.8

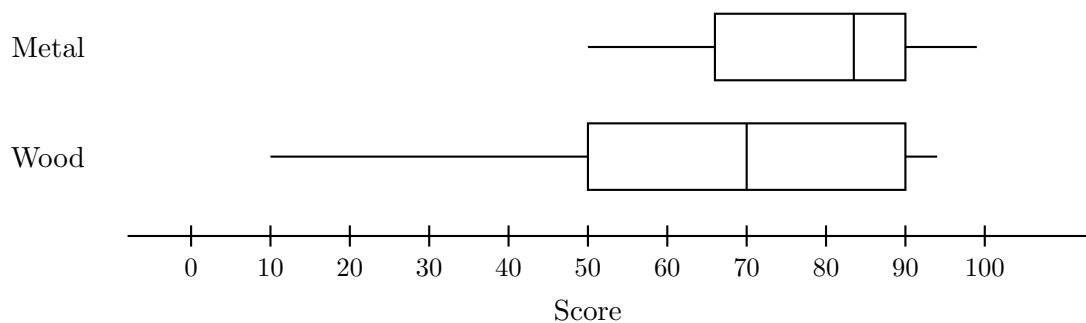
Girls:

50.5, 51.0, 51.4, 52.7, 52.8, 53.1, 53.6, 54.4, 54.4, 55.3

Use these data to draw side-by-side stem-and-leaf plots. Have your stems be 50, 51, 52, 53, 54, 55. Use $50 \mid 5 = 50.5$ as the legend.

Question 2 – Interpreting Boxplots

Every semester, an engineering professor assigns a project to her students to construct a catapult. Each catapult receives a score out of 100. The professor notices that catapults built out of metal tend to be of higher quality and earn higher scores than those built out of wood. Side-by-side boxplots were made with data she has collected over the years.



Question 3 – Basic Probability

When we roll two fair, independent six-sided dice (one blue and one red), we usually add the two numbers that appear. Suppose instead we pay attention to the maximum. For example, if the red die shows 3 and the blue die shows 4, then the maximum is 4. The table below has 36 cells corresponding to 36 equally likely outcomes. The maximum corresponding to each outcome appears in the appropriate cell. Answer problems a-d.

		<i>red die</i>					
		1	2	3	4	5	6
<i>blue die</i>	1	1	2	3	4	5	6
	2	2	2	3	4	5	6
	3	3	3	3	4	5	6
	4	4	4	4	4	5	6
	5	5	5	5	5	5	6
	6	6	6	6	6	6	6

- (a) What is $P(\max = 4)$?
- (b) What is $P(\{\max = 6\} \cup \{\text{red} = 1\})$?
- (c) What is $P(\max = 3 \mid \text{blue} = 3)$?
- (d) Are the events $\max = 1$ and $\text{red} = 1$ independent, mutually exclusive, or neither? Justify your answer.

Question 4: Expected value of Discrete R.V.

There is an unfair die with unusual numbers on its six faces. Let Y denote the outcome of one roll of the die. The probability distribution of Y is given below.

Y	$P(Y = y)$
4	0.30
6	0.10
9	0.10
11	0.10
15	0.20
30	0.20

Suppose that every time you roll this die, you win a number of dollars equal to the number that appears. There is a catch, however: you must pay **\$12 for each roll**.

Is Y a discrete or continuous random variable? Does it follow a geometric, binomial, or Poisson distribution, or none of these?

In the long run, do you expect to **win money**, **lose money**, or **break even**? Why? If you perform any calculations to answer this question, show your work.

Question 5 – Tree diagram, Bayes Rule and Conditional Probability

Bob's Bakery makes two kinds of pies: apple and blueberry. If you select a pie at random, the probability it is apple is 0.64. Bob's Bakery also makes two sizes of pies: large and small. The probability that a randomly selected apple pie is large is 0.70. The probability that a randomly selected blueberry pie is large is 0.60. Answer problems (a)–(b).

(a) What is the probability that a randomly selected pie is large?

(b) Given that a randomly selected pie is small, what is the probability that it is an apple pie?

Question 6 – Discrete R.V.

Balls are drawn at random, one at a time, from a basket containing 100 balls: 53 orange, 41 yellow, and 6 purple. Whether the draws are made with or without replacement is specified below.

For problems (a), (b), (e), and (f), identify the random variable as **binomial**, **geometric**, or **neither**. If it is binomial, identify the values of n and p . If it is geometric, identify the value of p .

- (a) Draw 5 balls **without replacement**. Let A be the number of purple balls drawn. Identify the distribution of A .

- (b) Draw 11 balls **with replacement**. Let B be the number of yellow balls drawn. Identify the distribution of B .

- (c) For the 11 draws with replacement described in part (b), find the probability that exactly 5 of the balls drawn are yellow. That is, find $P(B = 5)$.

Note: Show the appropriate probability formula with the given values substituted. You may leave your final answer in exact form; decimal approximations are not required.

- (d) Suppose you draw 4 balls at random, one at a time, **with replacement**. What is the probability that at least one of the 4 balls drawn is yellow? **Note:** Show the appropriate probability formula with the given values substituted. You may leave your final answer in exact form; decimal approximations are not required.

- (e) Draw balls **without replacement** until the first orange ball appears. Let C be the number of draws required. Identify the distribution of C .

- (f) Draw balls **with replacement** until the first purple ball appears. Let D be the number of draws required. Identify the distribution of D .

Question 7 – Discrete R.V.

WebRock, an online music site, has become very popular, and its server is overloaded an average of 3.2 times per day. Let Y be the number of times the server is overloaded in a day. Assume that Y follows a Poisson distribution with parameter $\lambda = 3.2$. Answer problems (a)–(b).

Note: Show the appropriate probability formula with the given values substituted. You may leave your final answer in exact form; decimal approximations are not required.

- (a) What is the probability that the server will be overloaded exactly 2 times tomorrow?

- (b) What is the probability that the server will be overloaded at least once tomorrow?

Question 8: Continuous R.V.

A scientist studying flu epidemics claims that the waiting time, in years, until the next flu outbreak in Disasterberg follows a uniform distribution on the interval $(0, 11)$. If the scientist is correct, what is the probability that the town will not suffer any flu outbreaks for the next 6 years?

Question 9 – Continuous R.V.

You decide to purchase a used truck from your friend Paul. Paul warns you that the truck is very old. He believes that the waiting time, in months, until the transmission fails follows an exponential distribution with rate parameter $\lambda = 0.05$ per month. Answer problems (a)–(c).

- (a) If Paul is correct, what is the probability that the truck's transmission will survive for at least the next 24 months?

- (b) What is the expected waiting time until the transmission fails?

- (c) Which discrete random variable is closely related with the exponential distribution? Can you define such a discrete random variable for this problem?

Question 10 – Joint Distribution: Discrete R.V.

Suppose X and Y have the following joint PMF and marginal PMFs:

$X \backslash Y$	0	1	2	$p_X(x)$
0	$\frac{2}{10}$	$\frac{1}{10}$	0	$\frac{3}{10}$
1	$\frac{1}{10}$	$\frac{2}{10}$	0	$\frac{3}{10}$
2	0	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{4}{10}$
$p_Y(y)$	$\frac{3}{10}$	$\frac{4}{10}$	$\frac{3}{10}$	1

Find $\text{Cov}(X, Y)$. Show your calculations for $E[X]$, $E[Y]$, and $E[XY]$.

Question 11 – Joint Distribution: Continuous R.V.

Suppose X and Y have the joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} \frac{2}{27}(x+y), & 0 < y < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Answer parts (a)–(b).

(a) Find the marginal probability density function of Y , $f_Y(y)$. Clearly specify its support.

(b) Find the conditional probability density function of X given $Y = 1$, $f_{X|Y}(x | 1)$. Clearly specify its support.